

SPC_20_R2

Variant: Prod

9th April 2026
R2

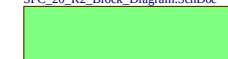
Release

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Blocks Smart Solar Charger
SPC_20_R2_TopPage.SchDoc



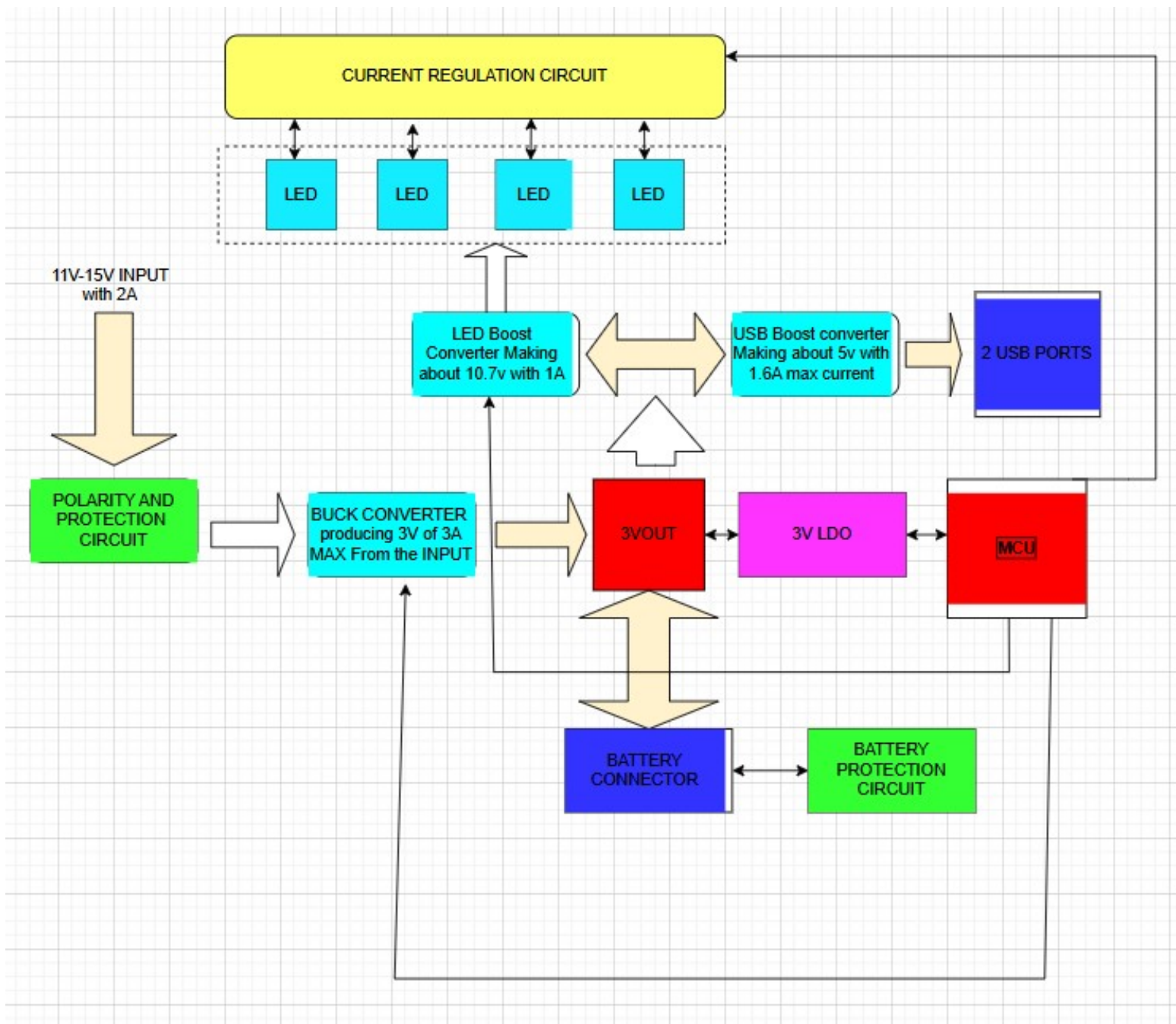
Block Diagram
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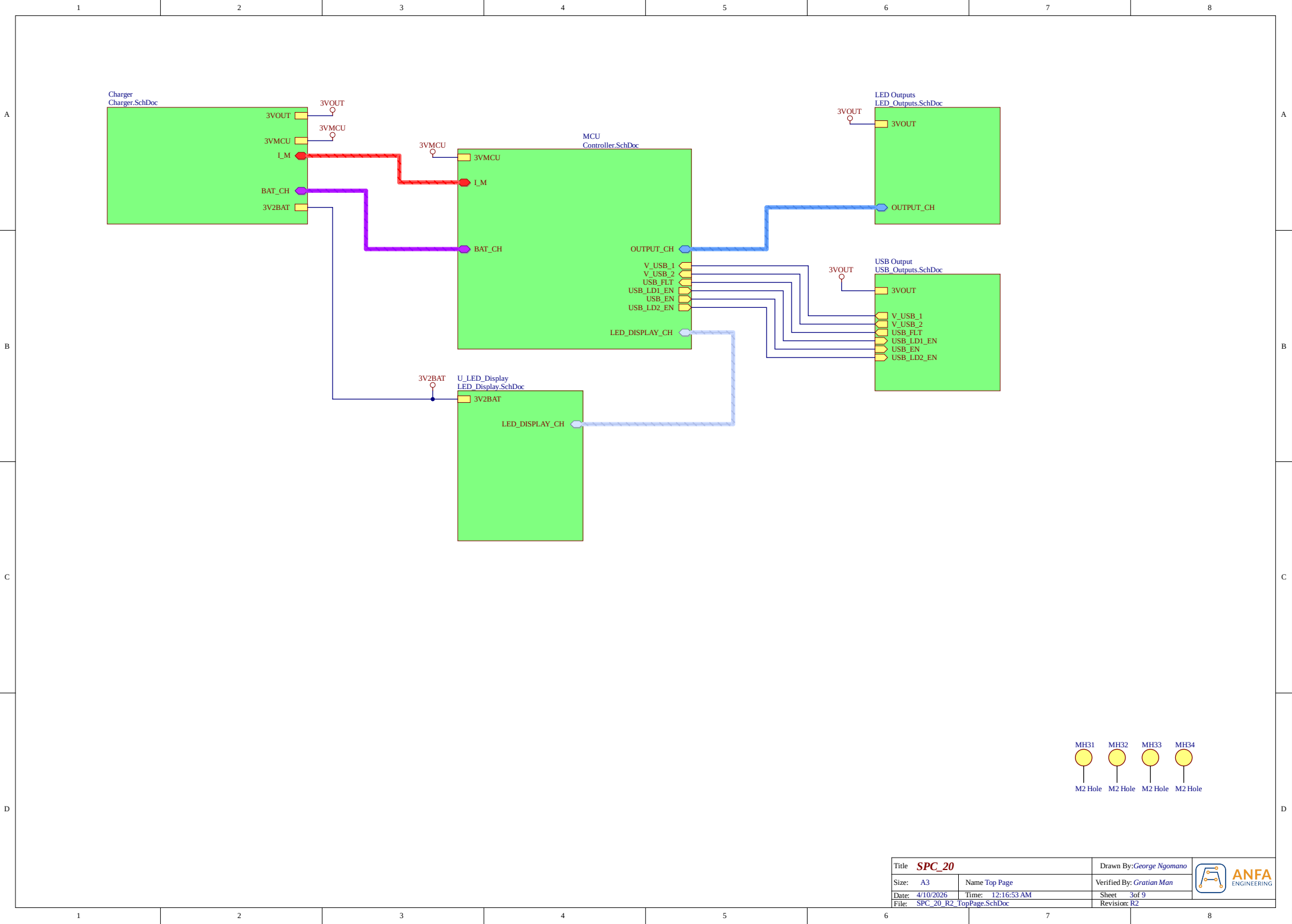


Changes
Changes.SchDoc



Title <i>SPC_20</i>		Drawn By: <i>George Ngomano</i>		
Size: <i>A3</i>	Name <i>Front Page</i>	Verified By: <i>Gratian Man</i>		
Date: <i>4/10/2026</i>	Time: <i>12:16:53 AM</i>	Sheet <i>1</i> of <i>9</i>		
File: <i>SPC_20_R2_Front.SchDoc</i>		Revision: <i>R2</i>		

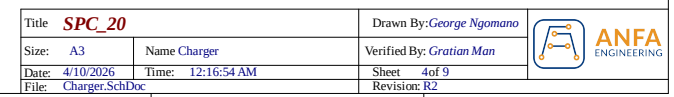


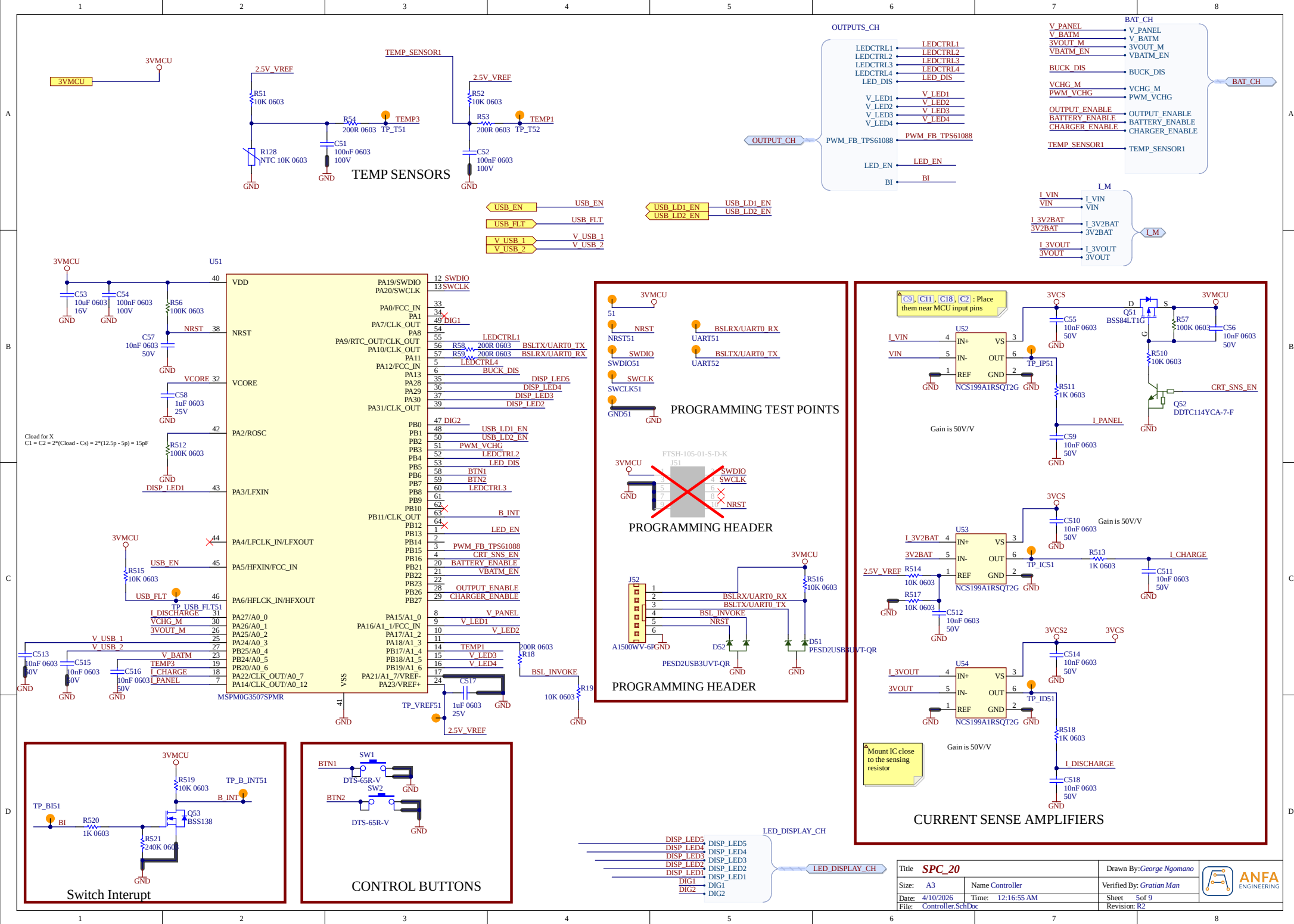


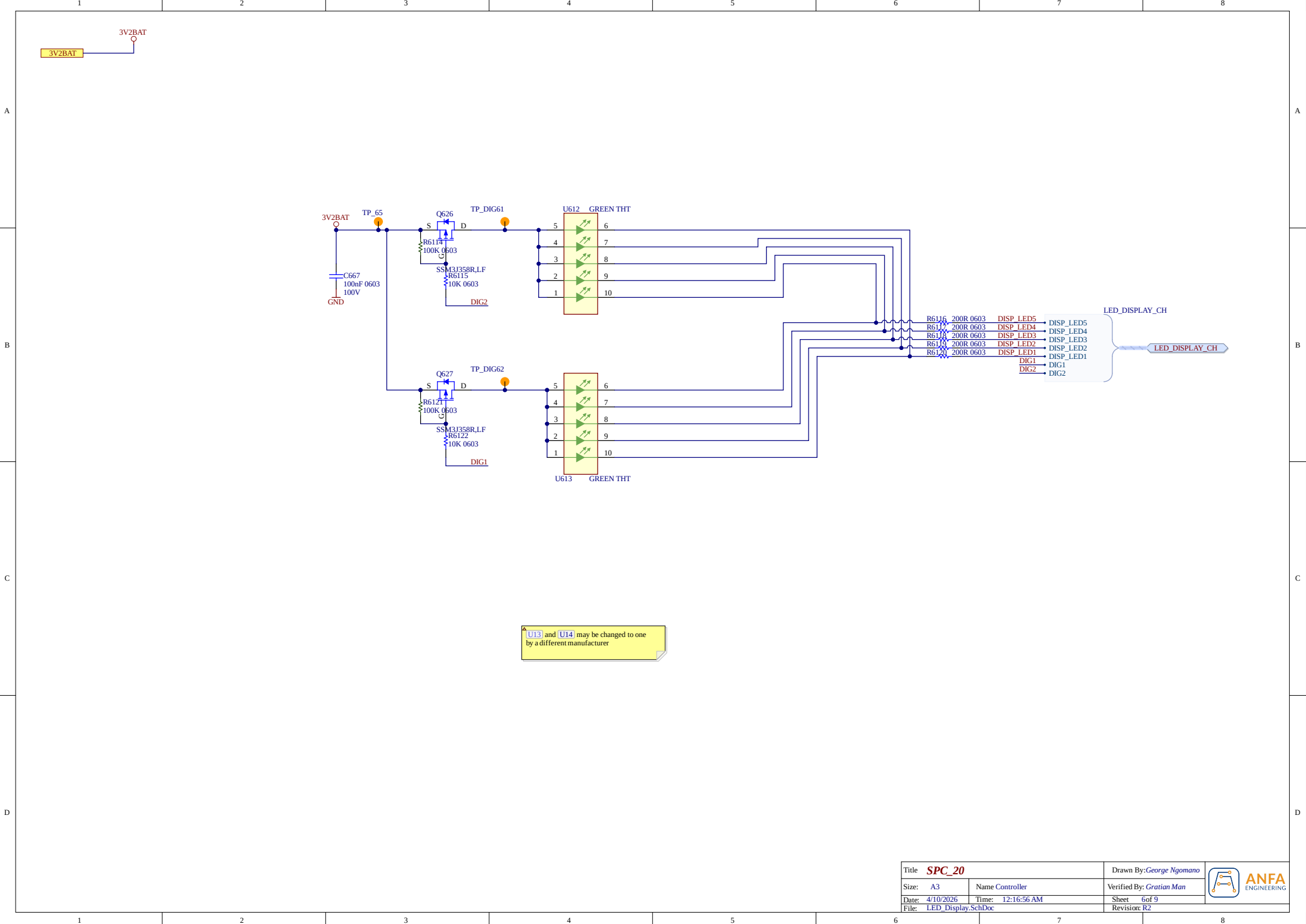
MH31 MH32 MH33 MH34
M2 Hole M2 Hole M2 Hole M2 Hole

Title SPC_20		Drawn By: George Ngomano	
Size: A3	Name Top Page	Verified By: Gratian Man	
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File: SPC_20_R2_TopPage.SchDoc		Revision: R2	









A

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C

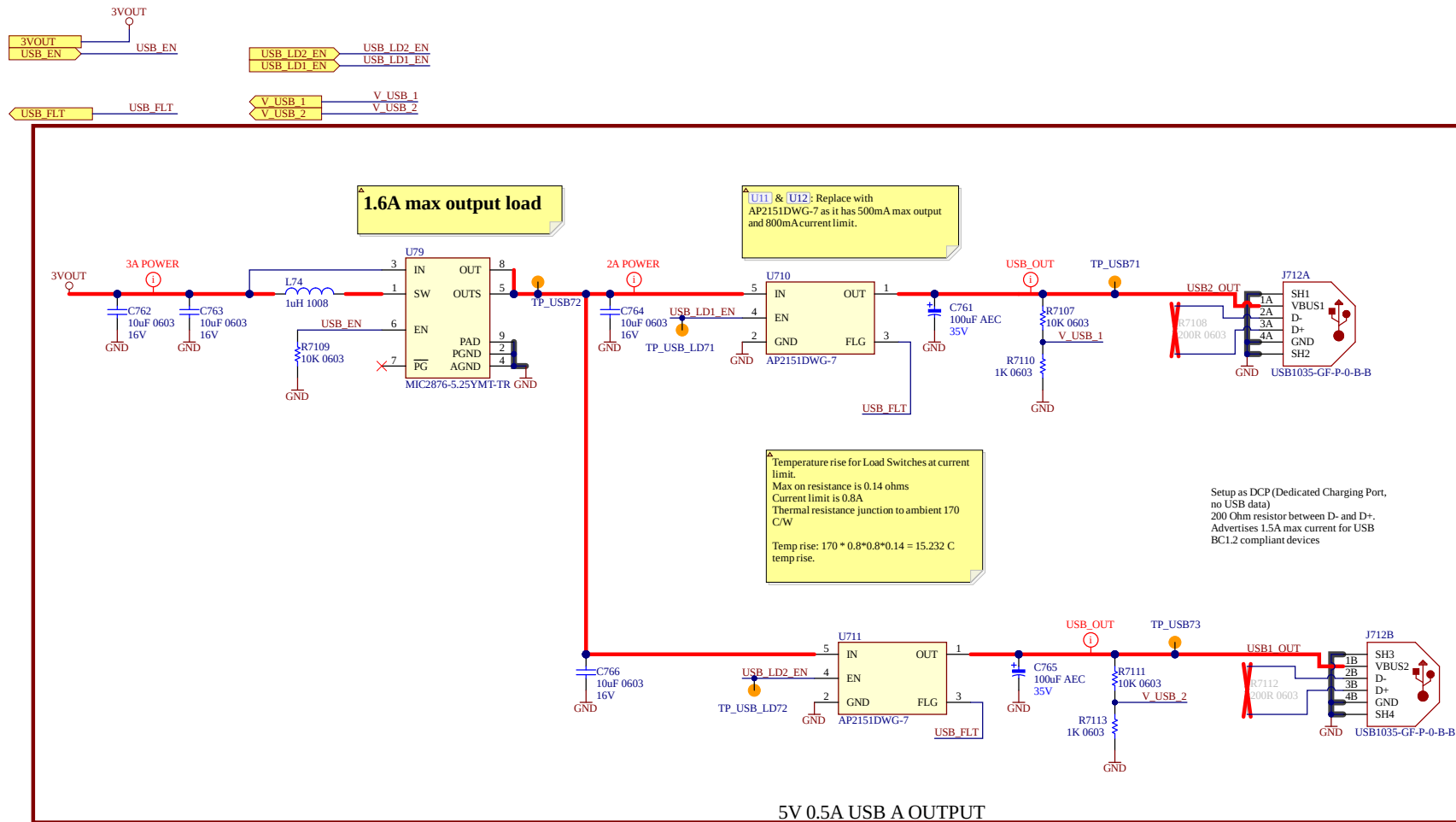
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A

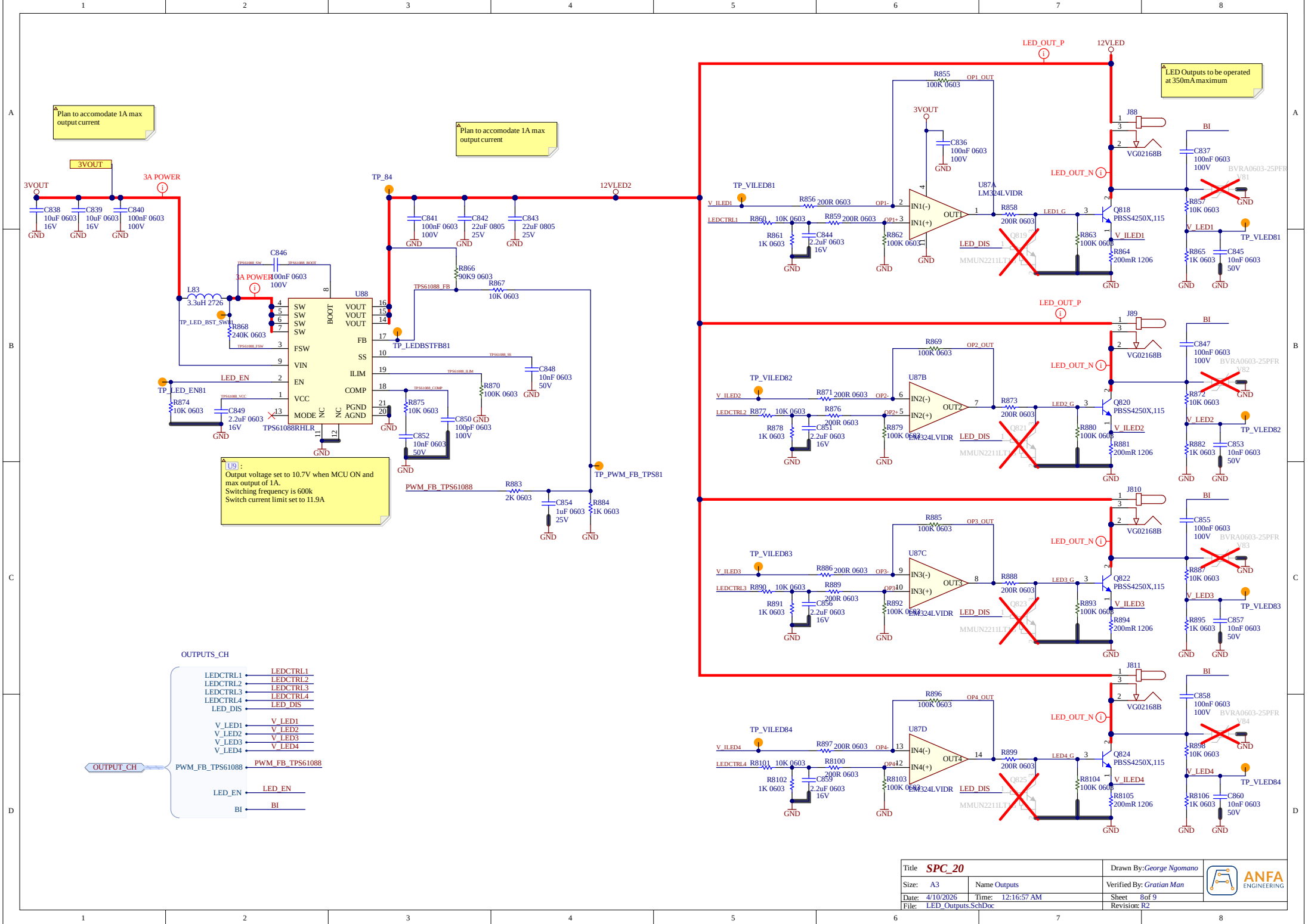
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D



Title SPC_20		Drawn By: <i>George Ngomano</i>	
Size: A3	Name USB Outputs	Verified By: <i>Gratian Man</i>	
Date: 4/10/2026	Time: 12:16:56 AM	Sheet 7 of 9	
File: USB_Outputs.SchDoc		Revision: R2	



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	Date		Change																																	
A	9.04.2026		1. Changed component designators to match ComponentDesignator_SheetNumber_ComponentIndex for easier location of which sheet components are on 2. Added two more LED outputs and changed OPAMP from LM358 to LM324 3. Updated MCU to LED Outputs Harness to reflect additions to LED Outputs 4. Changed the battery protection IC to BQ29706DSER 5. Changed the battery protection IC mosfets to a ECH8693R dual mosfet common drain: 6. Added the BSL signal to the MCU pin 7.Added the voltage regulator powered from VIN in parallel with D6. with assembly variant D6 or the woltage regulator. 8. Replaced the input filter inductor with a 1UH 1008 inductor. 9. Replace the buttons for the SPC20 with the PN: DTS-65R-V. (shorter compared to the first)																																	
B																																				
C																																				
D																																				
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